

Introduction

Natural Language Processing

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Course Flow (Tentative)

Introduction, Text Normalization (3)

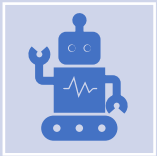
Embeddings, Classification (4)

Transformers, Foundation Models (4)

Generative AI (1)

Latest Research(2) + Presentations (2)

What is NLP?

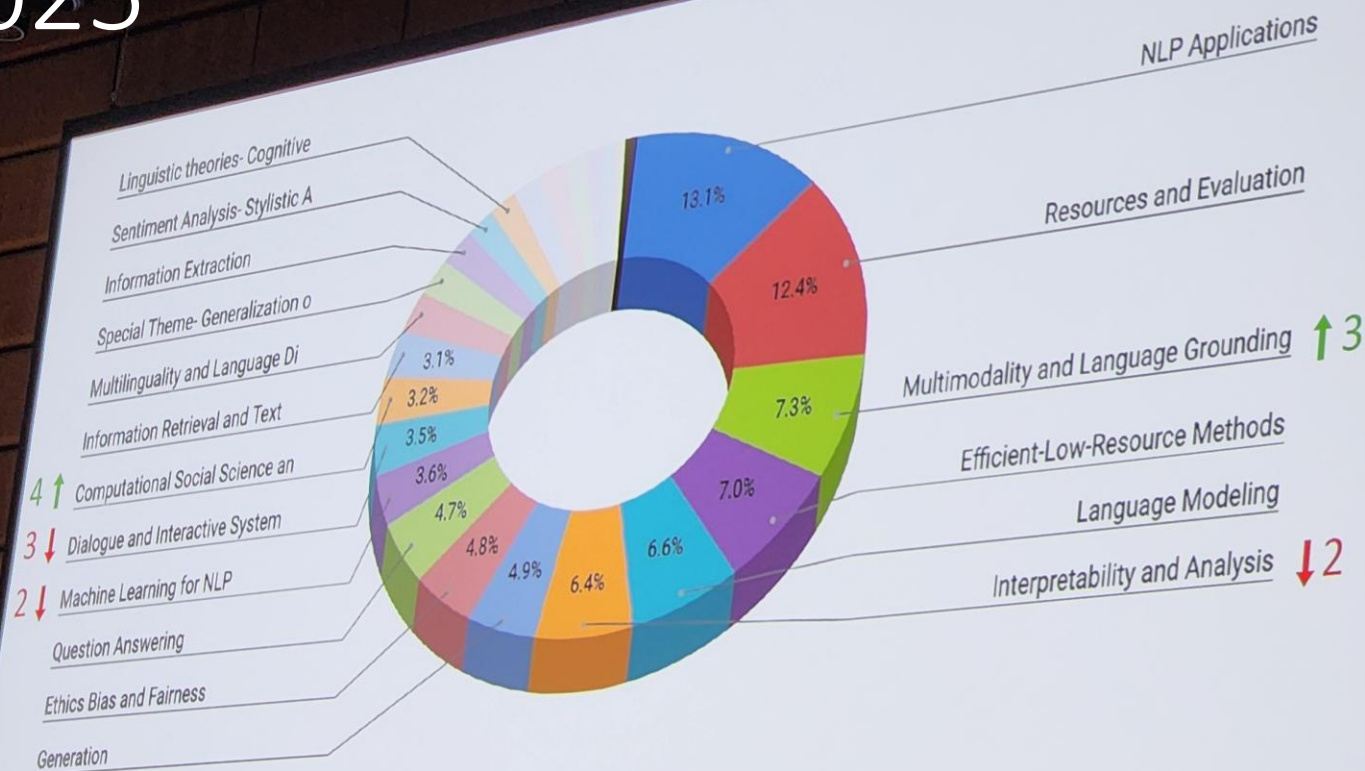


A branch of computer science—and more specifically, the branch of artificial intelligence or AI—concerned with giving computers the ability to understand text and spoken words in much the same way human beings can (IBM).



An interdisciplinary subfield of linguistics, computer science, and artificial intelligence concerned with the interactions between computers and human language, in particular how to program computers to process and analyze large amounts of natural language data (Wikipedia).

ACL 2025



- 2024: NLP Applications, Resources and Evaluation, Efficient/Low-Resource, Interpretability, Generation, Multimodality, ML, Dialogue, Ethics, QA, IE
- 2023: NLP Applications, ML, IE
- 2022: ML, IE, NLP Applications
- 2021: ML, MT & Multilinguality, IE

Before Transformers

A Brief History of Chatbots

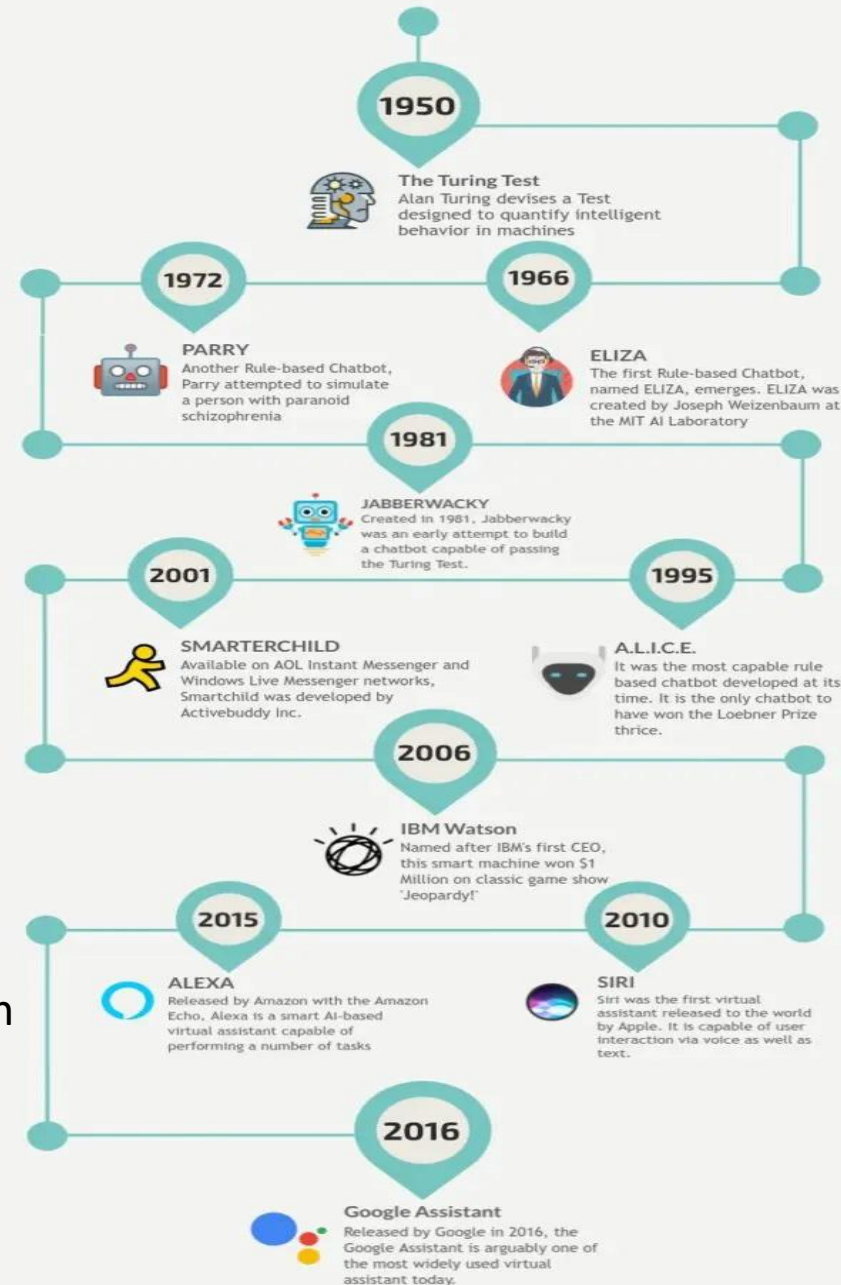
Stanford

MIT

Richard Wallace, Artificial Linguistic
Internet Computer Entity

Amazon

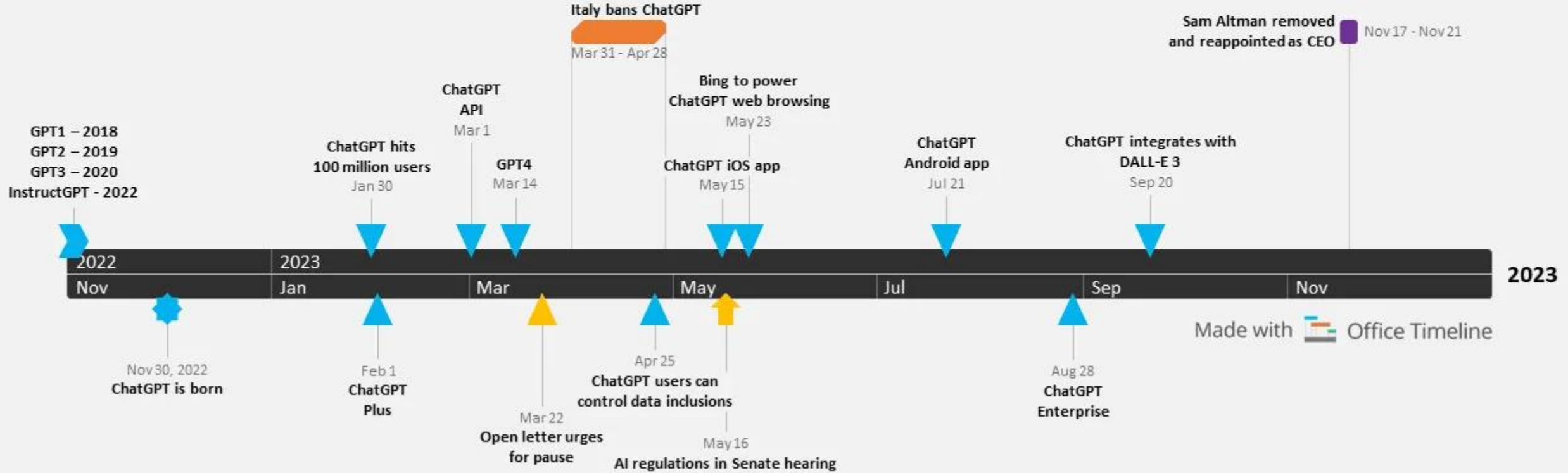
Apple



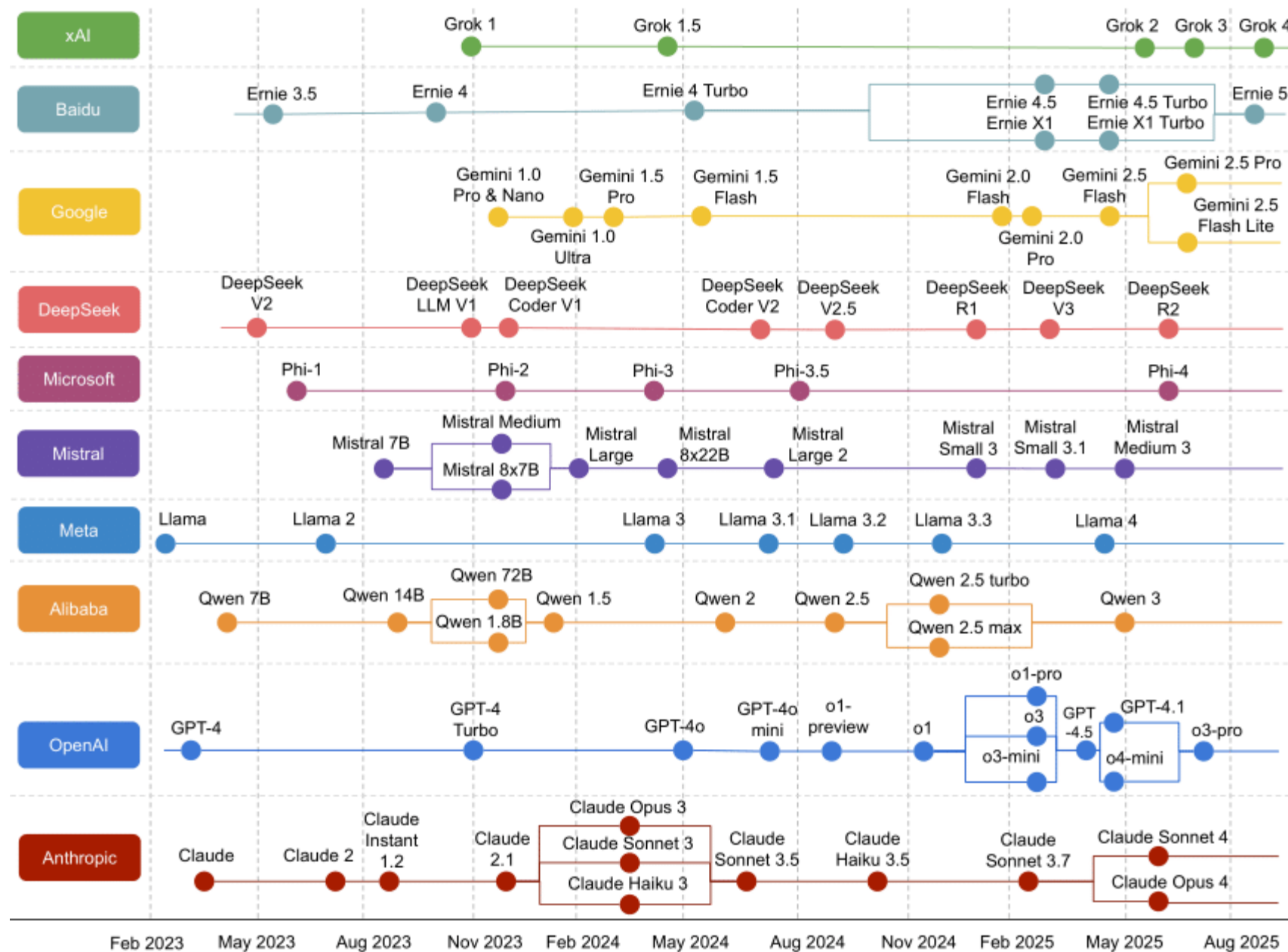
Reference:
<https://datasciencedojo.com/blog/what-are-chatbots/>



ChatGPT history and timeline



- https://www.researchgate.net/figure/Timeline-of-major-LLM-releases-2023-Early-2025-showing-the-rapid-evolution-of-LLMs-fig3_393983430
- <https://llmtimeline.web.app/>



Aspect	High-End Models	Small/Efficient Models	Open-Source	Specialized Models
Leaders	OpenAI, Anthropic, Google, xAI	Microsoft (Phi), Mistral, Google (Nano)	Meta (LLaMA), Mistral	DeepSeek (Coder), Google (Flash), Qwen
Strategy	Premium performance & reasoning	Cost-effective deployment	Research accessibility	Domain/task-specific optimizations
Cadence	~3–6 months between major updates	Faster iteration possible	Steady, research-driven	Rapid specialization cycles

Capabilities of the Most Popular AI Tools

Best  Yes  No 

	 ChatGPT	 Claude	 Gemini	 Grok	 perplexity
Everyday answers					
Writing					
Coding					
Reasoning					
Deep research					
Web search					
Voice chat					
Image gen					
Video gen					
Live camera					
Computer use					

<https://creatoreconomy.so/p/chatgpt-vs-claude-vs-gemini-the-best-ai-model-for-each-use-case-2025>

Year 2024

- A claim that GPT-4 passed a Turing test ([see Reference](#)).

ChatGPT response:

As of now,, ChatGPT has **not officially passed** the Turing Test. The Turing Test, originally proposed by Alan Turing in 1950, is a measure of a machine's ability to exhibit intelligent behavior indistinguishable from that of a human. While ChatGPT and other advanced AI models have demonstrated impressive conversational abilities, they have not been formally tested or recognized as having passed the Turing Test in a standardized or widely accepted manner.

The Turing Test itself has also been a topic of debate and reinterpretation over the years, with some arguing that it may no longer be a sufficient measure of true artificial intelligence.

Note: It could give you a different response, try yourself.

Do we need a new criteria to define Artificial Intelligence?

- Commonsense
- General Knowledge
- Social intelligence
- Post Turing: Mapping the landscape of LLM Evaluation (<https://arxiv.org/abs/2311.02049>)
- On the Measure of Intelligence <https://arxiv.org/abs/1911.01547>
- Are Large Language Models Intelligent? Are Humans? <https://www.mdpi.com/2813-0324/8/1/68>

Other Benchmarks

- Story Cloze Test:
 - This test requires a system to choose the correct ending to a four-sentence story.
 - Commonsense test
- MMLU (Massive Multitask Language Understanding):
 - The benchmark to measure knowledge acquired during pretraining by evaluating models exclusively in zero-shot and few-shot settings.
 - A massive multitask test consisting of multiple-choice questions from various branches of knowledge.
 - <https://arxiv.org/pdf/2009.03300>
 - Many more...

Applications

- Education
- Health
- Management
- Marketing

Mental Health Analysis

- Counseling conversation analysis ([Althoff et al., 2016](#))
- Synthetic data generation from mental health datasets for NLP ([Ive et al., 2020](#))
- Report summarization ([Shah et al., 2021](#))
- Chatbot to support or identify mental health issues ([Denecke et al., 2020](#))
- Social media analysis to identify population-level symptoms of anxiety, depression, and suicide risk ([Low et al., 2020](#); [Fine et al., 2020](#))

Education

- Intelligent Tutoring Systems
- Automated Essay Scoring or Grading
- Plagiarism Detection
- Text Summarization
- Mental Health

Marketing

- Ad Targeting and Personalization
- Customer Support (Chatbots & Virtual Assistants)
- Customer Feedback Analysis

Management

- Decision Support Systems
 - Risk Management
- Meeting Summarization
 - Transcription
- Hiring
 - Resume Screening

Challenges

- **Sentence 1:** *"On a cold, **fall** day, Alex went to the **bank** by the river to deposit his paycheck. The **fall** season was his favorite time of year. After finishing his errand, Alex sat by the river **bank** and watched the leaves **fall** from the trees."*
- **Sentence 2:** *"He had just turned in his assignment on time, despite it being due on **10/12/2023**. His professor, however, follows the European date format, and thus expected it by **12/10/2023**. To avoid confusion in the future, Alex decided to confirm the due dates in both **MM/DD/YYYY** and **DD/MM/YYYY** formats."*

Word Sense Disambiguation (WSD):

- **Word "fall":**

- First occurrence: *"On a cold, fall day"* (Season - autumn)
- Second occurrence: *"watched the leaves fall from the trees"* (Verb - to drop)

- **Word "bank":**

- First occurrence: *"went to the bank by the river"* (Financial institution)
- Second occurrence: *"sat by the river bank"* (Side of the river)

Part-of-Speech (POS) Tagging:

- **"Fall"** in different contexts:
 - *"fall day"* (Adjective - describing the day)
 - *"leaves fall"* (Verb - describing the action of dropping)
- **"Bank"** in different contexts:
 - *"bank by the river"* (Noun - referring to the side of the river)
 - *"bank to deposit"* (Noun - referring to a financial institution)

Varying Date Formats:

- **10/12/2023**: Interpreted as October 12th in the U.S. format, but as December 10th in the European format.
- **12/10/2023**: Interpreted as December 10th in the U.S. format, but as October 12th in the European format.

How to identify Varying date Formats?

Some solutions

Suppose an NLP tool encounters the date "10/12/2023" in a text identified as English. The tool might:

- Check for language and locale clues (e.g., U.S. vs. U.K. English).
- Look for other dates in the text to determine if a pattern is evident.
- Use regex to identify the date format and apply context to resolve ambiguity.
- If the text mentions "October 12th, 2023" explicitly elsewhere, it might use this to infer that "10/12/2023" follows MM/DD/YYYY.

Challenges (Date Formats)

- **Cultural Differences:** Different countries and regions use different formats, and text can contain a mix of these.
- **Ambiguity:** Some dates are inherently ambiguous, like "03/04/2023," which could be March 4th or April 3rd depending on the format.

NLP tasks

- Part-of-Speech tagging
- Word-sense disambiguation
- Named Entity Recognition
- Sentiment Analysis
- Summarization
- Machine Translation
- Question Answering
- Text Generation

Grading

- **Readings (5%):** Will be uploaded on the Canvas at least 2-3 days before the class.
 - Quiz or class discussion
- **Class Participation (5%):** Attendance + Positive class participation
- **Project (30%):** Implementation +Report +Demo
 - Group of 2/3
- **Presentation (5%):** Latest Research
 - Group of 2/3

- **Assignment (20%):**
 - 4 assignments (3 coding (individual)+ 1 Review paper/Report (Group))
- **Quizzes (20%)**
- **Mid-Exam (15%)**

Project Ideas(with prior ML experience)

- <https://www.aclweb.org/portal/content/eaclacl-2026-joint-call-workshops>

Important Dates

EACL/ACL 2026 shared dates:

- Proposal submission deadline: **September 5, 2025**
- Notification of acceptance: September 22, 2025

All deadlines are 11:59PM UTC-12:00 (“anywhere on Earth”).

- <https://semeval.github.io/SemEval2026/cft>

Preliminary timetable

- Sample data ready 15 July 2025
- Training data ready 1 September 2025
- Evaluation data ready 1 December 2025 (internal deadline; not for public release)
- Evaluation start 10 January 2026
- Evaluation end by 31 January 2026 (latest date; task organizers may choose an earlier date)
- Paper submission due February 2026
- Notification to authors March 2026
- Camera ready due April 2026
- SemEval workshop Summer 2026 (co-located with a major NLP conference)

- <https://lrec2026.info/>